PH PHEWPHONICS · MAD FRAMEWORK

MAD RUN — OCTOBER 2026 · AA · DA · SSA

Flight-Stress-Assassin-Plus Findings

Multi-Agent Debate covering flight data aggregation architecture, API viability, bot-detection realities, ToS risk, and a phased build plan. Born from a real AI failure: bad advice to "wait for a sale" from an AI that couldn't access live flight data.

● CONDITIONAL BUILD — 3 PHASES

FLIGHT SEARCH | AA · DA · SSA | AMADEUS API | CLAUDE-IN-CHROME | OCTOBER 2026

SECTION 1

SSA Verdict

Final synthesised output from the Multi-Agent Debate cycle

SSA FINAL VERDICT

Conditional Build — Three-Phase Approach Required

The goal of "never stress about flights again" is achievable — but not by building what you first imagine. The AA is right that the technical tools exist. The DA is right that full automation is a fragile, legally grey maintenance trap. Build a hybrid: legitimate APIs where data is clean, browser assist where it's not, and never pretend to do what only a human can do at booking time.

3	7	5	1
BUILD PHASES	FAILURE MODES	CRITICAL BLOCKERS	DAY TO MVP

AA — TOP WINS

- Amadeus Self-Service API is free, production-grade, covers 400+ airlines — no scraping needed for full-service carriers
- Authenticated browser session bypasses bot detection entirely — use the user's own Chrome, not a headless bot
- Price calendar query finds cheapest day in a month automatically — core stress eliminator
- Phase 1 (tab launcher) can be built in one day with zero cost and zero legal risk

DA — TOP RISKS

- Bot detection is industrial-grade — Cloudflare/PerimeterX fingerprints canvas, WebGL, TLS, mouse entropy — headless browsers detectable in seconds
- Low-cost carriers (Jetstar, AirAsia, Scoot) are NOT in GDS — the cheapest fares are structurally inaccessible via API
- Dynamic pricing: prices change every 30–90 seconds — any scraped price is a snapshot, never a guarantee
- Booking is always manual — tool eliminates search stress, not all stress

SECTION 2

Debate Scorecard

AA vs DA — what the MAD cycle settled on each key question

QUESTION	AA SAID	DA SAID	SSA RULING
Can we get real flight data?	Yes — Amadeus API, SerpApi, 400+ airlines	Amadeus misses LCCs; data goes stale in 30–90s	PARTIAL *Amadeus for full-service; Google Flights deep-link for LCCs*
Does authenticated browser automation work?	Yes — user's own session bypasses bot detection entirely	Fragile on site updates; ToS violation risk; sessions expire with 2FA	SPARINGLY *Claude-in-Chrome for LCC checks only — not as primary data pipeline*
Will this eliminate flight stress?	Yes — one query fans out to all sources	Booking still manual; price still unverified until click-through	REFRAME *Eliminates search stress (80%). Booking stress remains (20%)*
Is full scraping legally safe?	With user's own session, minimal risk	ToS explicitly prohibits; loyalty account ban risk; CFAA exposure	STAY CLEAN *Google Flights deep-link + Amadeus = zero ToS exposure*
Is Skyscanner API available?	Kiwi.com Tequila API as alternative	Skyscanner API closed to new partners since 2019	BLOCKED *Use Amadeus + Travelpayouts instead; Kiwi Tequila where available*
Can LCC fares be aggregated?	Via Claude-in-Chrome authenticated session	Jetstar/AirAsia/Scoot not in GDS; must scrape direct	MANUAL BRIDGE *Google Flights deep-link covers most LCCs; Chrome automation as fallback*

SECTION 3

Three-Phase Build Plan

Build in this order. Phase 1 alone eliminates most search stress.

01

THE LAUNCHER

Build time: 1 day · Cost: $0

Input form takes origin, destination, date range, stops, baggage, time windows, airlines. Generates pre-filtered deep-links for Google Flights, Skyscanner, Qantas, Virgin, Jetstar. One click opens all in parallel tabs. Zero scraping. Zero ToS risk. Zero maintenance. Eliminates re-entering the same data into five sites every time.

02

THE INTELLIGENCE LAYER

Build time: 1 week · Cost: $0/month (free tier)

Amadeus Self-Service API adds: cheapest day in a date range, price trend signal (is today high or low vs 90 days?), seat availability alerts, threshold price notifications. Dashboard shows all active searches and alerts in one view. Covers full-service carriers (Qantas, Emirates, Singapore, Virgin). LCCs handled by Phase 1 launcher.

03

THE LCC BRIDGE

Build when needed · Optional

Claude-in-Chrome attaches to user's existing authenticated Jetstar/AirAsia session. Manual trigger only — not scheduled. Extracts prices, merges into dashboard alongside Amadeus data. Used sparingly for LCC routes not covered by Amadeus. Treat as a research signal, not a booking source — always verify on site.

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FLIGHT-STRESS-ASSASSIN-Plus Architecture
│
├── LAUNCHER (Phase 1 — Day 1)
│   ├── Input: origin, dest, dates, stops, baggage, time windows, airlines
│   ├── Generates: Google Flights + Skyscanner + Qantas + Virgin + Jetstar deep-links
│   └── → Opens all in parallel tabs · Zero cost · Zero ToS risk
│
├── INTELLIGENCE (Phase 2 — Week 1)
│   ├── Amadeus API: price calendar · cheapest day · trend signal · availability alert
│   ├── Dashboard: all active searches + price alerts in one view
│   └── → Free tier: 2,000 calls/month · Covers 400+ full-service airlines
│
└── LCC BRIDGE (Phase 3 — Optional)
    ├── Claude-in-Chrome: user's existing Jetstar/AirAsia session
    ├── Manual trigger only · Results merged into dashboard
    └── → Research signal only — always verify on airline site before booking
```

SECTION 4

7 Critical Failure Modes to Avoid

Identified by the DA. None are caught by standard development planning.

1. **Don't scrape without user session.** Cloudflare/Akamai fingerprints canvas rendering, WebGL, TLS hash, mouse entropy. A headless Playwright session is detectable within seconds. Will be blocked within days. Use Amadeus API or the user's own authenticated Chrome session instead.

2. **Don't trust scraped prices as bookable.** Prices change every 30–90 seconds. Any price shown is a snapshot at time of query. Always verify on the airline site before any purchase. Build a "verify on airline site" button into the UI — make it the default next step.

3. **Don't try to cover all airlines on day one.** Start with Amadeus (400+ full-service carriers) + Google Flights deep-link (covers LCCs via the site). Add Claude-in-Chrome LCC bridge incrementally when a specific gap is felt. Trying to cover everything at launch means shipping nothing.

4. **Don't build a background scheduling service for scraping.** Scheduled automated sessions on airline sites = ToS violation. Loyalty account termination risk (points gone). Use on-demand search only. Price alerts via Amadeus API are fine — that's using an authorised data channel.

5. **Don't attempt to automate the booking step.** Seat selection, payment, confirmation are always manual and always on the airline site. The tool's job is search aggregation and handoff — not end-to-end booking. Attempting booking automation introduces payment failure modes and zero benefit.

6. **Don't conflate "price found" with "deal confirmed."** The UI must always show data freshness (timestamp of last query) and include a prominent "verify before buying" warning on every result. A user who acts on a stale price will blame the tool, not the airline's dynamic pricing system.

7. **Don't let "AI said so" replace the tool.** The origin story of this project is an AI that gave bad flight advice because it had no real data. This tool must never give a price recommendation without showing its data source, its freshness, and its limitations. Data transparency is non-negotiable.

SECTION 5

Honest Delivery Assessment

What this tool will and won't deliver. No false promises.

CAPABILITY	DELIVERED?	VIA
Never re-enter the same search parameters again	✓ Yes	Phase 1 Launcher — day one
Cheapest day in a date range found automatically	✓ Yes	Amadeus price calendar API — Phase 2
Know if today's price is high or low vs recent history	✓ Yes	Amadeus + Travelpayouts price history — Phase 2
Alert when price drops to threshold	✓ Yes	Amadeus subscription API — Phase 2
One click opens all relevant sites pre-filtered	✓ Yes	Phase 1 Launcher — day one
LCC fares (Jetstar, AirAsia, Scoot) surfaced	✓ Mostly	Google Flights deep-link (Phase 1) + Chrome bridge (Phase 3)
Qantas/Velocity Points fares surfaced	× Not yet	No public API; future Phase 3+ enhancement
Booking handled by the tool	× No	Always manual on airline/OTA site — by design
Guaranteed bookable price shown	× No	Dynamic pricing means verification always required
Seat selection, baggage fine print, check-in links	× No	Not aggregatable — handled on airline site at booking

NET ASSESSMENT

Eliminates 80% of the Friction. Build Phase 1 This Week.

Phase 1 alone (the launcher — one day of build time) removes the single biggest source of flight search stress: re-entering the same parameters into five different sites. Phase 2 adds the intelligence layer that the original AI advice failed to provide — real price trend data, not a guess. Phase 3 fills the LCC gap when needed. The remaining 20% — booking, seat selection, price verification — cannot be automated without unacceptable legal and fragility risk. That's not a failure. That's an honest scope.